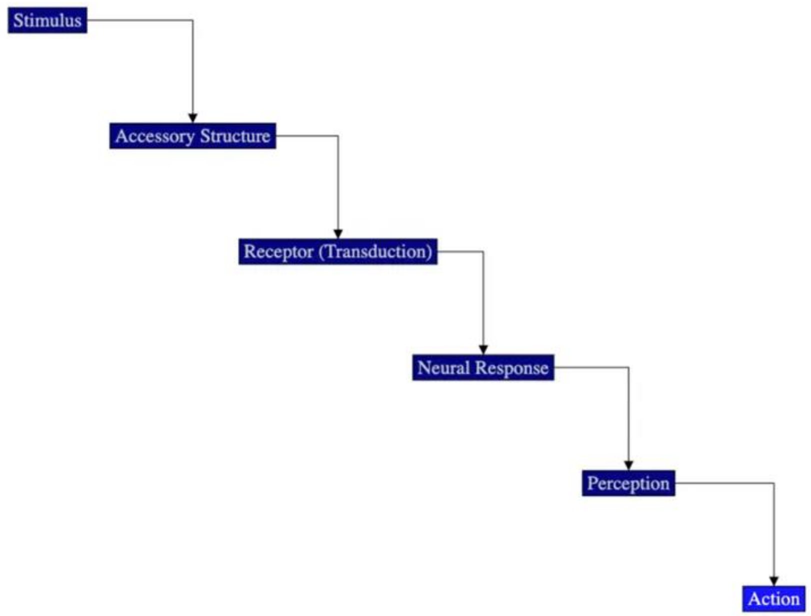
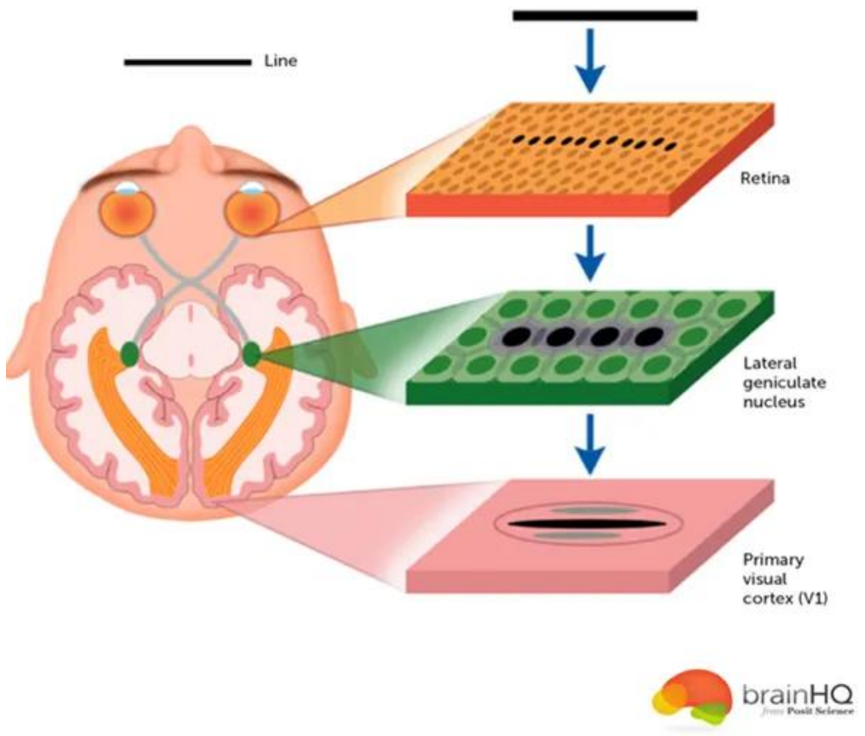
# Foundations

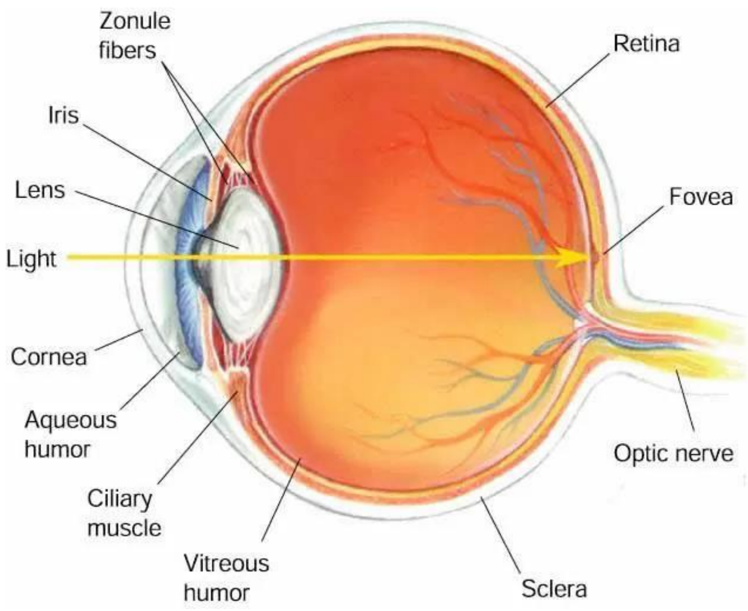
# Steps of sensory processing

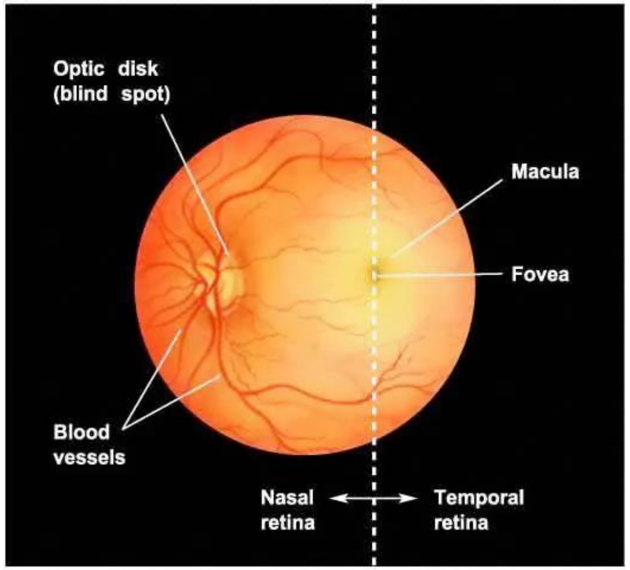


# The Eye:

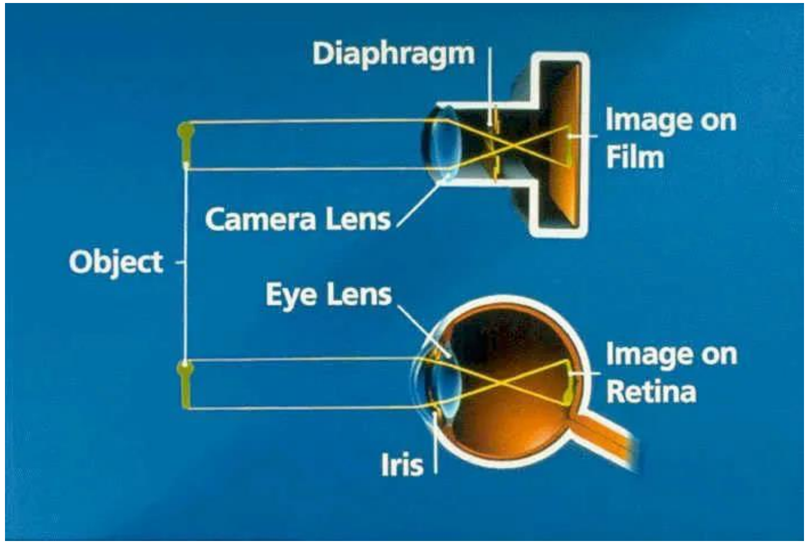
In a “simple” or “camera eye”,
•The **iris** adjusts to let in more or less light through the **pupil**.
•the image is inverted on the retina.
•The blood vessels that supply the intraretinal circulation enter and leave the eye at the **optic disk**. The optic disk is also the place where optic axons leave the eye and become the **optic nerve**. The optic disk has no receptors and is thus a **blind spot**.

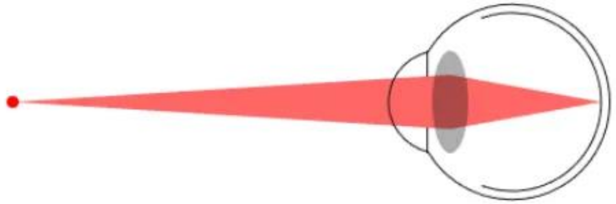


# The Lens: Accommodation

- The Lens is the adjustable focusing element of the eye.
- It is located just behind the iris.
- This process of adjusting the focus for different distances by changing the shape of the lens is called accommodation.
- This process is very rapid although changing accommodation from a near object to a far object is faster than going from a far object to a near object
- Accommodation is controlled by muscles connected to the lens, called ciliary muscles.
    - The ciliary muscles work automatically without conscious control.
    - They can contract and increase the curvature of the lens so that the lens thickens.
- The increased curvature of the lens allows the eye to focus on a close object.
    - When the person then has to look at a faraway object, the muscles relax and the focus of the lens changes to an object further away.



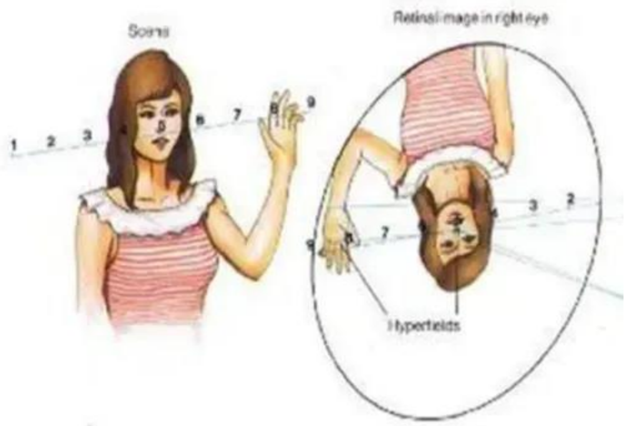



https://isle.hanover.edu/Ch03Eye/Ch03Accommodation.html

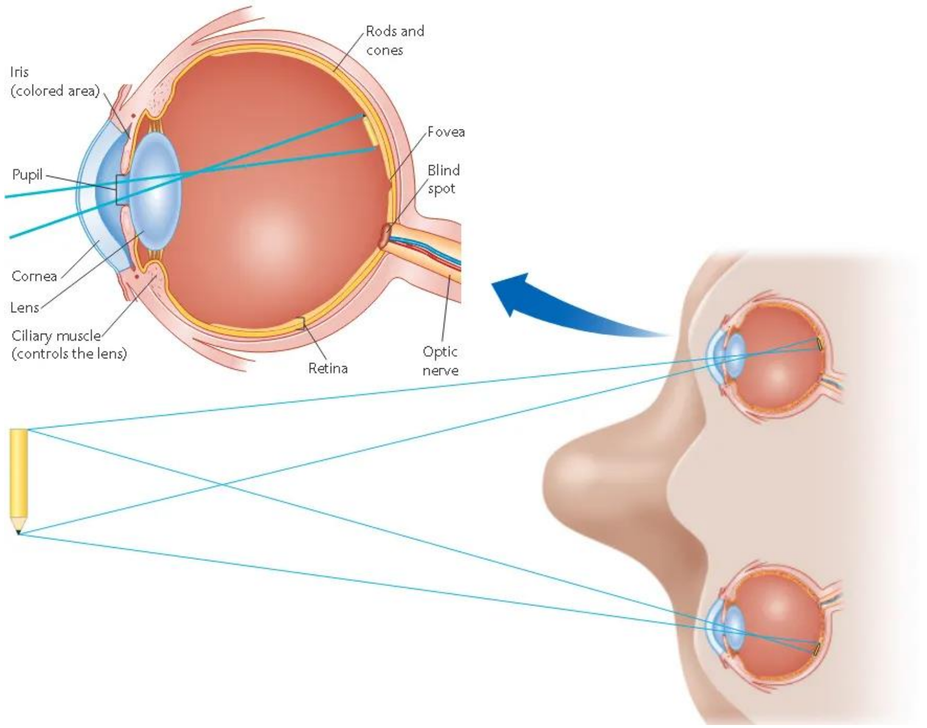
# The Lens: Image inversion

Everything gets reversed when it passes through the lens

An object in the visual field produces an inverted image on the retina

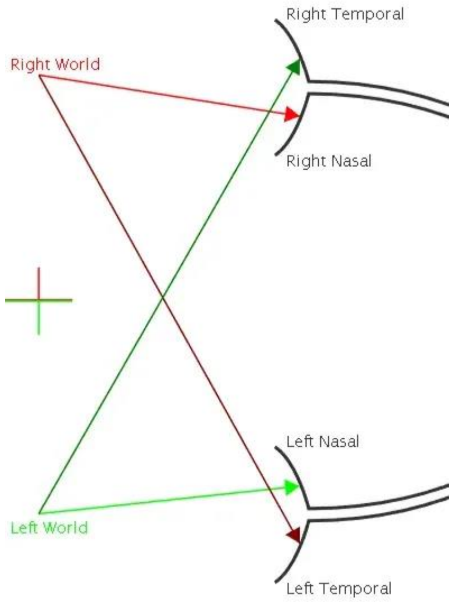


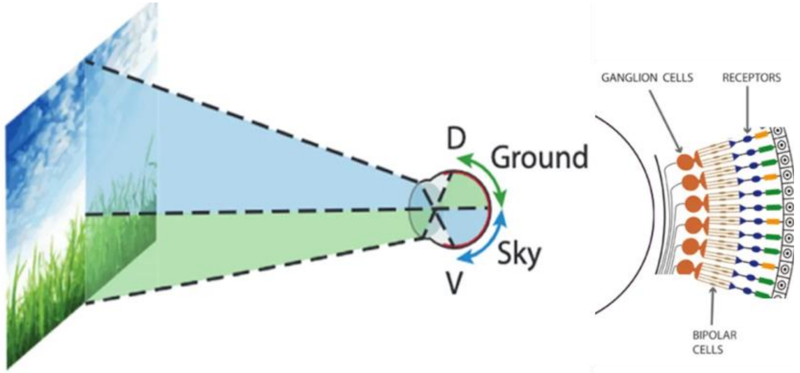
An object in the visual field produces an inverted image on the retina. The optic nerve exits the eyeball on the nasal side (the side closer to the nose).



# Image Projection onto the Retina

Transformation of the image by the lens has consequences for the retina...



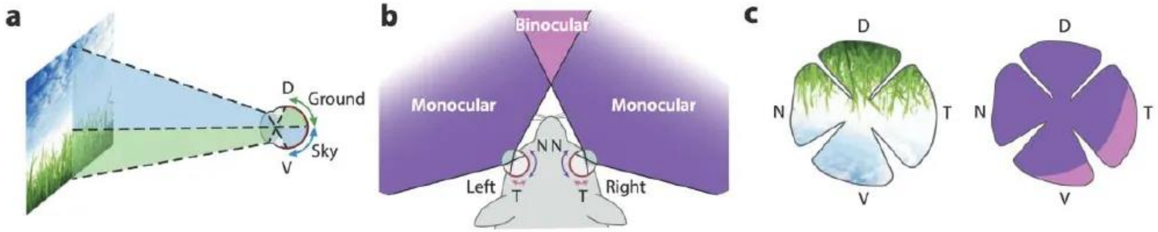
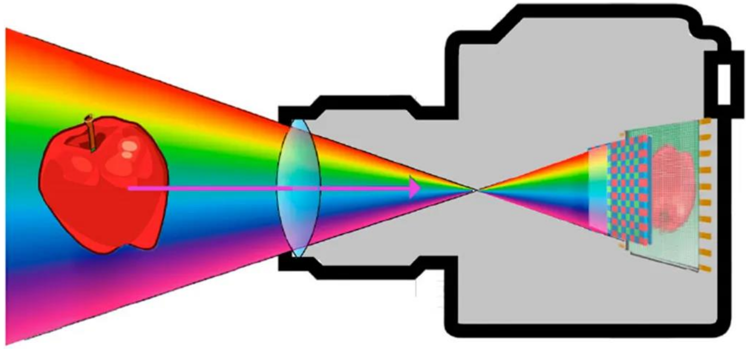
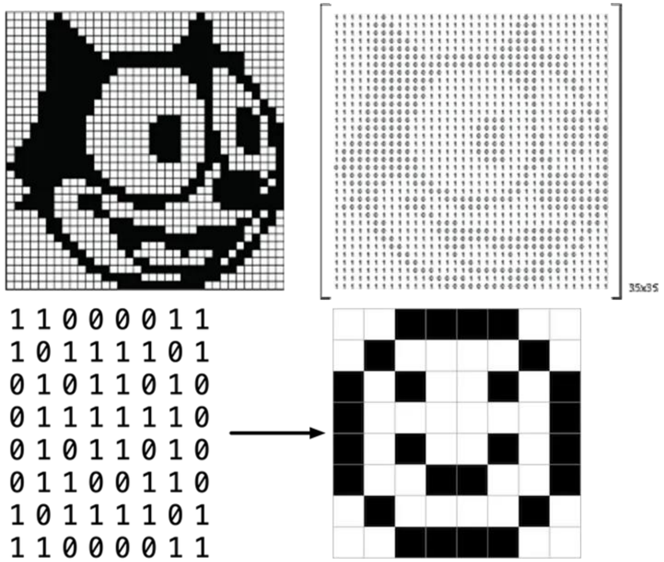




**Figure 1 Visual space of the mouse**

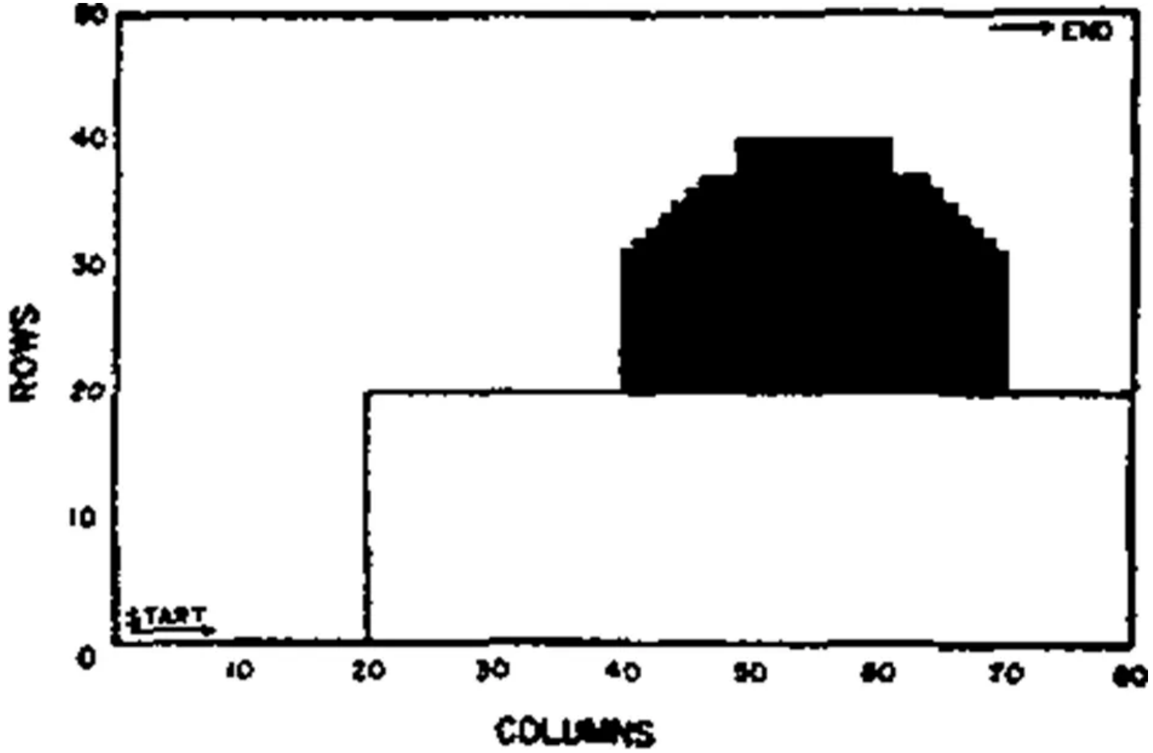
*(A) Images from the upper and lower part of visual space project onto the ventral and dorsal regions of the mouse retina, respectively.* ***(B)*** *The central region of visual space projects onto the temporal retinas of both eyes, while images from the lateral environment project onto the nasal retinas.* ***(C, left)*** *Example of an image of the mouse's environment projecting onto its retina.* ***(C, right)*** *Illustration of the retinal regions receiving information from the central visual field (pink regions) and the lateral visual field (dark purple regions); Figure adapted from Heukamp 2020*

# Images can be described as a set of pixels





# A note on Information content



# THE RETINA

Three layers of cells and two layers of synapses.

<u>Optic nerve fibers</u>

Ganglion cell layer

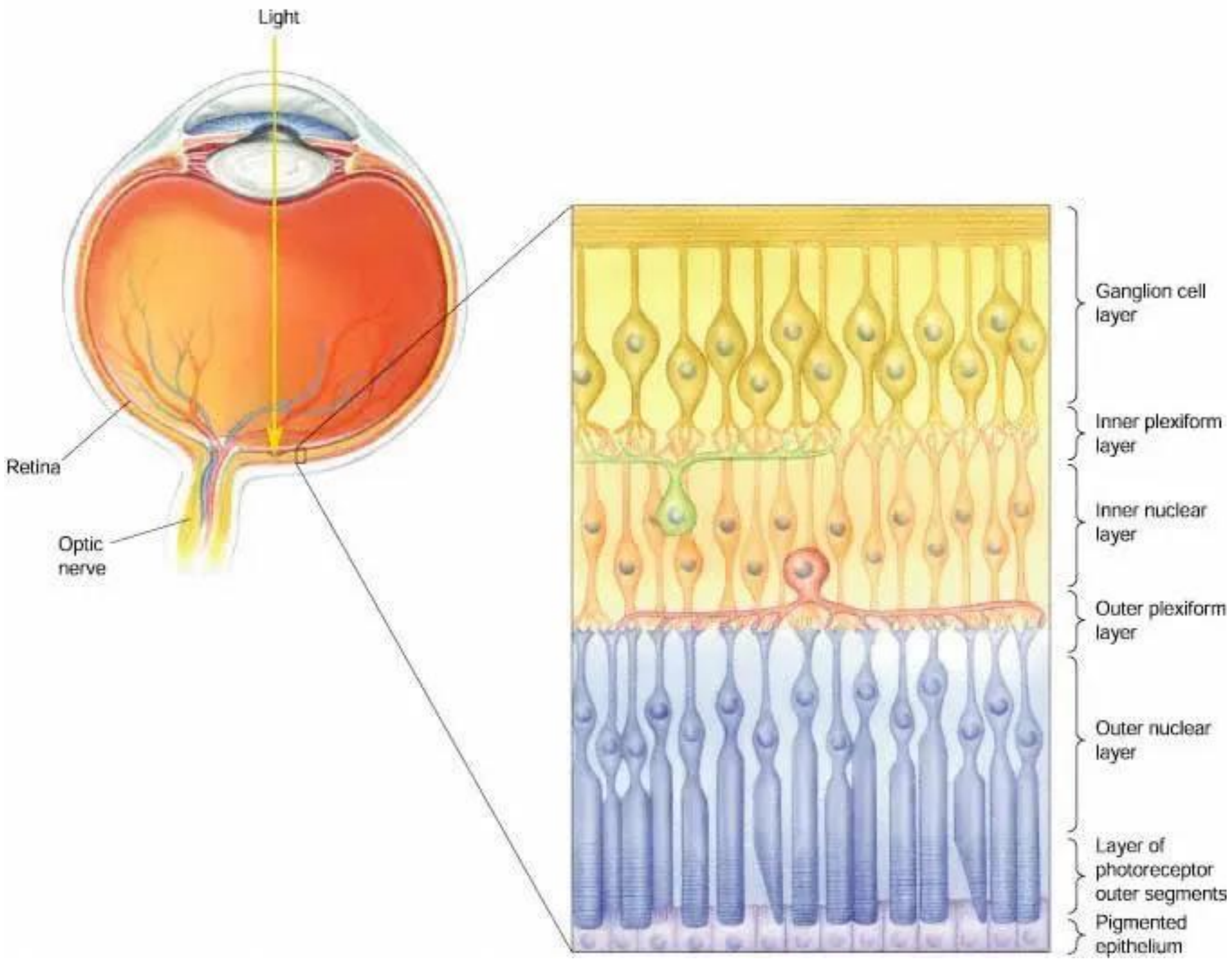
<u>Inner plexiform layer (IPL)</u>

Bipolar cell layer: *horizontal cells* + *bipolar cells* + *amacrine cells*

<u>Outer plexiform layer (OPL)</u>

Photoreceptor cells: rods & cones

Pigmented epithelium



# THE RETINA

Photoreceptors

Rods work at low (scotopic) light levels.

Cones work at high (photopic) levels and are responsible for both sharp acuity and color vision.

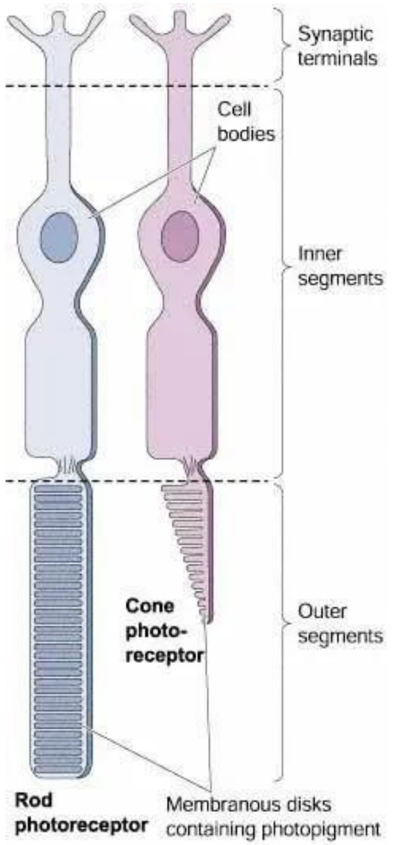
Both work at intermediate (mesopic) light levels.

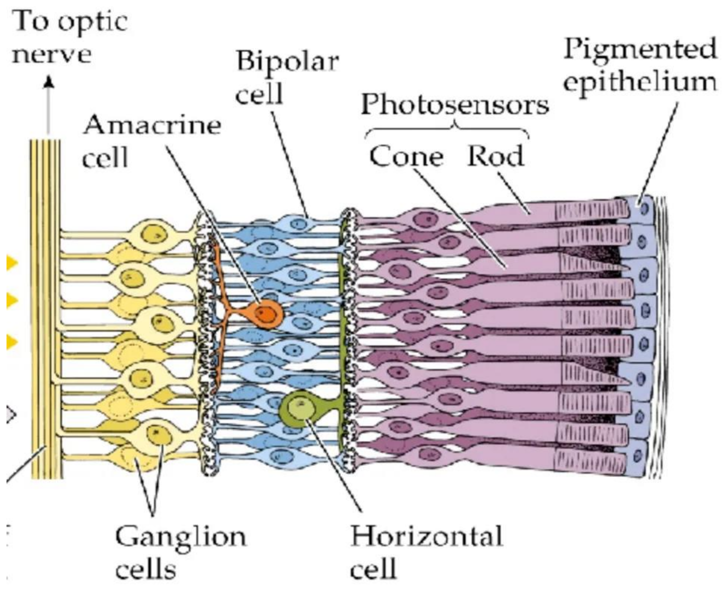
The direct pathway is from cones → bipolar cells → ganglion cells.

Horizontal cells and amacrine cells provide lateral connections. H cells do NOT contact bipolar cells.

Cones and bipolar cells give graded responses.

Ganglion cells fire spikes.



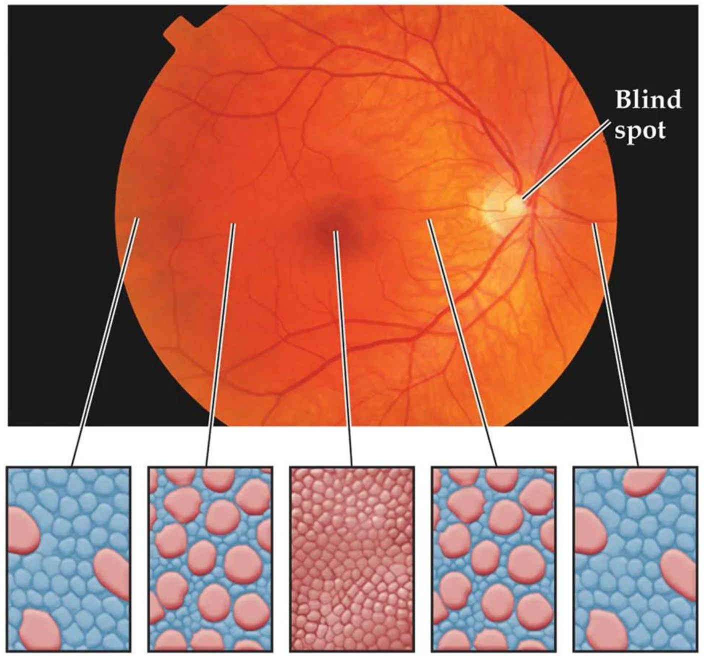
# Photoreceptor Layout

**Layout (**Humans):
the center (fovea) has only cones
Visual acuity is best in the **fovea**–it has a high density of cones.

**Optic disc**–where blood vessels enter and leave the eye and where "optic axons" from ganglion cells leave the eye.

**Blind spot** due to lack of photoreceptors in the optic disc

*(a)* Distributions of rods and cones across the retina

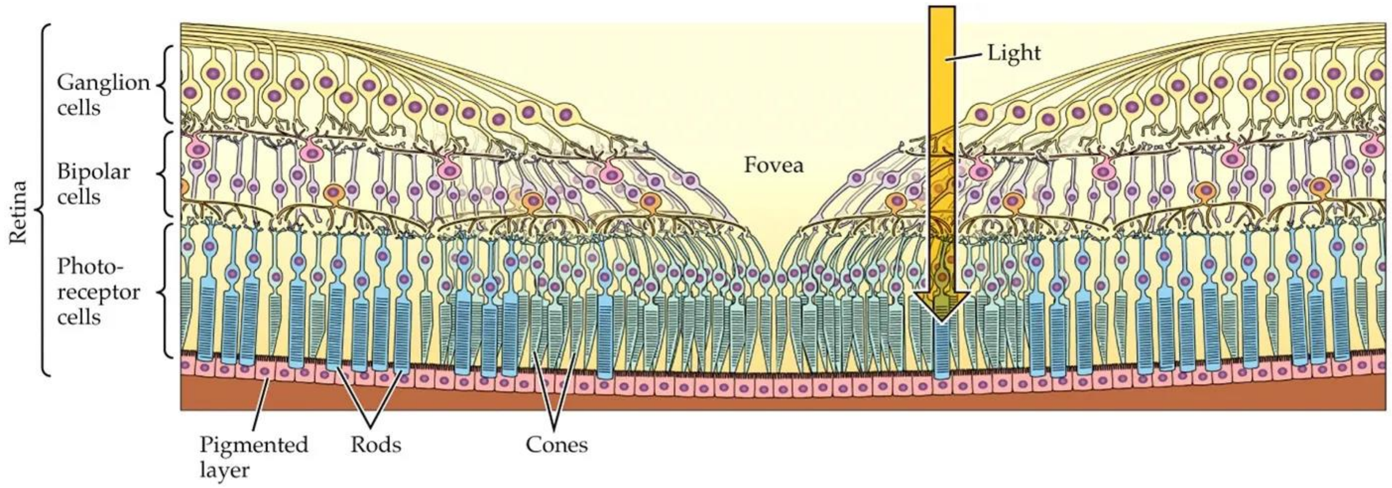


# The Fovea

**The fovea**
The fovea ('pit'), at the center of the retina, has no rods, only densely packed cones.

The cone terminals, bipolar cells and ganglion cells, along with their intraretinal blood supply, are swept to the side, so the path of light to foveal cones is unimpeded by blood.

Clinically, the center of the retina is called the macula. Macular degeneration is the most common cause of blindness in the elderly.

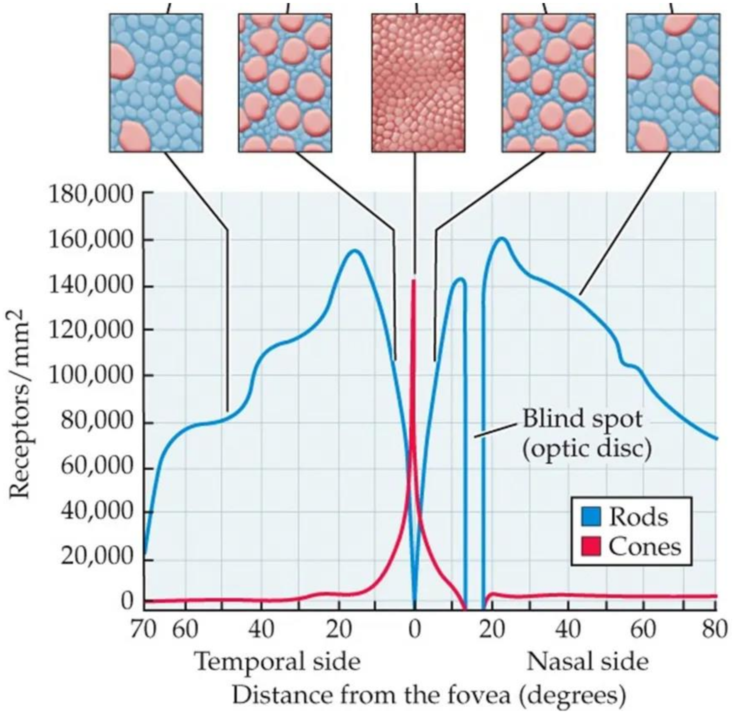


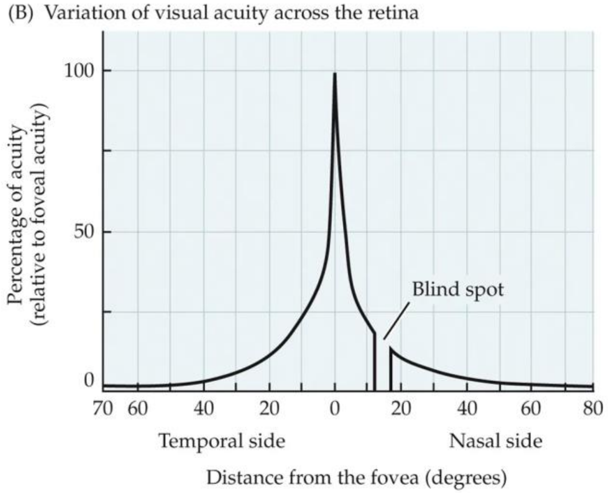
# The Fovea

**Visual resolution in the fovea**
Foveal cones are very densely packed, giving just ~2.5 mm spacing. With 300 mm/deg of visual angle in human, this is ~1/120th deg. With 120 samples/deg, our theoretical best resolution 60 cycles/deg. Remarkably, our resolution limit is this good!

Therefore, the circuits driven by foveal cones must be able to preserve the sampling density of the cones. The midget bipolar/midget ganglion cell pathway does this.

Resolution follows ganglion cell spacing, which increases away from the fovea because
- a) the density of cones falls away from the fovea,
- b) more and more cones converge on each midget bipolar cell, and
- c) more and more midget bipolar cells converge on each midget ganglion cell.




*BIOLOGICAL PSYCHOLOGY 7e*, Figure 10.5 (Part 2)


[The Blind Spot: https://michaelbach.de/ot/cog-blindSpot/](https://michaelbach.de/ot/cog-blindSpot/)

# Readings + Assignment

**Readings/Assignments (MUST DO)**
Submit your research question on friday
Read and Send me a paper on visual prosthetics. Once I confirm we'll both read

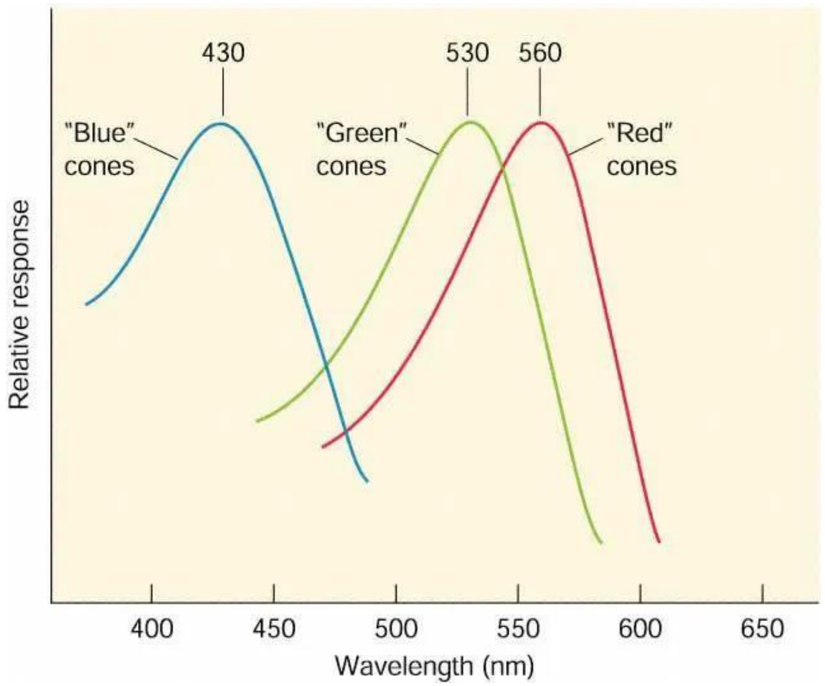
**Read Chapter 12 The Nervous System and Nervous Tissue (MUST READ)**
https://openstax.org/books/anatomy-and-physiology/pages/12-introduction 12.1 Basic Structure and Function of the Nervous System 12.2 Nervous Tissue 12.3 The Function of Nervous Tissue 12.4 The Action Potential 12.5 Communication Between Neurons
From : J. Gordon Betts, Kelly A. Young, James A. Wise, Eddie Johnson, Brandon Poe, Dean H. Kruse, Oksana Korol, Jody E. Johnson, Mark Womble, Peter DeSaix (Book URL: https://openstax.org/books/anatomy-and-physiology/pages/1-introduction)

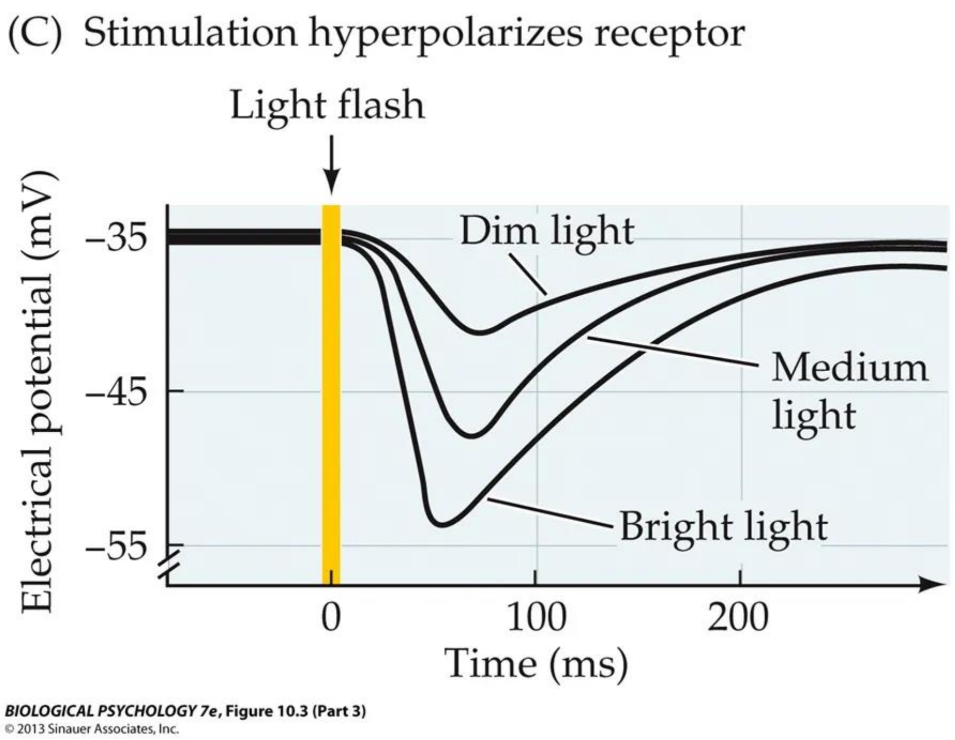
**Kalat JW (2024) Biological psychology, 14th edition. Boston, MA: Cengage. (MUST READ)**
- Module 5.1: Visual coding (stop reading at color vision)
- Module 5.2: Visual Processing in the Brain
- Module 5,3: Specialized Visual Processes

**Sensation and Perception Cengage Free chapter 1 (MAYBE READ)**
- Chapter 1. Introduction to Perception
  - 1.3. The Perceptual Proces
  - 1.4. Studying the Perceptual Process
  - 1.5. Measuring Perception
  - 1.6. Something to Consider: Why Is the Difference Between Physical and Perceptual Important?

**Next Meeting**
- retinal cell signaling
  - must have read up on action potentials
- central processing
- discuss paper you sent

# Next Week


430
530
560
"Blue" cones
"Green" cones
"Red" cones
Relative response
400
450
500
550
600
650
Wavelength (nm)


(C) Stimulation hyperpolarizes receptor
Light flash
Electrical potential (mV)
–35
–45
–55
Dim light
Medium light
Bright light
0
100
200
Time (ms)
BIOLOGICAL PSYCHOLOGY 7e, Figure 10.3 (Part 3)
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